SE444: Introduction to Artificial Intelligence
Mid-Term II Fall 2025

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Date: Monday, November 10, 2025 Time: 5:30 PM – 7:00 PM (90 minutes)

Instructions

- This is a closed-book exam. The University's code of ethics applies.
- Answer each exam question to the best of your knowledge.
- Clearly state any assumptions made that might be needed to understand your solution.
- Show your work – any sign of serious effort will be considered.
- No marks will be given for not making any effort.

Student Information

Student ID:	Student Name:

Summary of Exam Questions

Part	Topic	Time	Score	Mark	Mapped CLOs
Part I Multiple Choice	FOL — Empty Domain & Entailment	5 min	1		CLO 1
	Propositional Logic — Consistency	5 min	1		CLO 1
	CSP — Arc Consistency	5 min	1		CLO 1, CLO 2
	Logical Agents — Architecture	5 min	1		CLO 1, CLO 3
Part II Logical Reasoning	Translation to FOL	10 min	3		CLO 1
	Translation with Negation	10 min	3		CLO 1
	Negation and CNF Conversion	20 min	4		CLO 1
Part III CSP	Formulate Lab Scheduling as CSP	10 min	2		CLO 2
	Apply Arc Consistency (AC-3)	15 min	3		CLO 2
	Find a Consistent Solution	5 min	1		CLO 2
Total for Part I (Multiple Choice)		20 min	4		
Total for Part II (Logical Reasoning)		40 min	10		
Total for Part III (CSP)		30 min	6		
Total Points		90 min	20		

Part I. Critical-Thinking Multiple Choice Questions

Total Score: 4 points **Total Time:** 20 minutes

Instructions: Each question may have **one or more correct answers**. Select **all that apply**. Partial credit will be awarded for partially correct answers.

Answer Key Summary

Option	Q1 (5m)	Q2 (5m)	Q3 (5m)	Q4 (5m)
A	**X**		**X**	
B			**X**	**X**
C	**X**	**X**	**X**	
D				**X**
E			**X**	

Correct Answers:

- **Q1 (FOL):** A, C
- **Q2 (Propositional Logic):** C
- **Q3 (CSP):** A, B, C, E
- **Q4 (Agents):** B, D

Question I.1: First-Order Logic (FOL) — Quantifiers and Entailment

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

You are given the following two FOL sentences (assume a non-empty domain):

1. $\forall x\,(P(x) \rightarrow Q(x))$
2. $\exists x\,P(x)$

Which of the following sentences are logically entailed by (1) and (2)?

A) $\exists x\,Q(x)$

B) $\forall x\,Q(x)$

C) $\exists x\,(P(x) \wedge Q(x))$

D) $\forall x\,(Q(x) \rightarrow P(x))$

E) $\exists x\,(Q(x) \wedge \neg P(x))$

Answer: A and C

Explanation: Assuming a non-empty domain, (1) $\forall x\,(P(x) \rightarrow Q(x))$ and (2) $\exists x\,P(x)$ together allow us to infer the existence of at least one object that satisfies both P and Q. Hence, $\exists x\,Q(x)$ and $\exists x\,(P(x) \wedge Q(x))$ are both logically entailed.

Details:

A) **Correct.** From (2), there exists some a such that $P(a)$ holds. By (1), $P(a) \rightarrow Q(a)$, so $Q(a)$ must

also hold. Therefore, $\exists x\, Q(x)$ follows.

B) **Incorrect.** The premises do not guarantee that *every* x satisfies $Q(x)$—only that those satisfying $P(x)$ must also satisfy $Q(x)$.

C) **Correct.** From (2), there exists an a with $P(a)$. By (1), $Q(a)$ also holds for the same a. Thus, $\exists x\,(P(x) \land Q(x))$ is true.

D) **Incorrect.** (1) expresses $\forall x\,(P(x) \to Q(x))$, not the converse. The reverse implication $\forall x\,(Q(x) \to P(x))$ does not follow.

E) **Incorrect.** There is no information ensuring the existence of an element that satisfies $Q(x)$ but not $P(x)$.

Key Concept: Combining an existential statement $\exists x\, P(x)$ with a universal conditional $\forall x\,(P(x) \to Q(x))$ allows us to infer both $\exists x\, Q(x)$ and $\exists x\,(P(x) \land Q(x))$, but not universal claims about all x.

Question I.2: Propositional Logic — Consistency and Entailment

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1

Consider a knowledge base (KB) with propositional atoms p, q, r, s, t:

1. $(p \land q) \to r$
2. $r \to (s \lor t)$
3. $\neg s \land \neg t$
4. q
5. $p \lor \neg r$

Which of the following statements is true?

A) KB is unsatisfiable.

B) KB $\models p$.

C) KB $\models \neg p$.

D) KB $\models (r \land s)$.

E) KB can be satisfied with $p = \text{true}$ and $r = \text{true}$.

Answer: C

Explanation: The KB is satisfiable but forces p to be false through a chain of reasoning.

Step-by-step analysis:

Step 1: From (3), we know $\neg s \land \neg t$, so both s and t are false.

Step 2: From (2), $r \to (s \lor t)$. Since s and t are both false, $(s \lor t)$ is false. For the implication to be true, r must be false (because true $\to$ false would be false).

Step 3: From (4), we know q is true.

Step 4: From (1), $(p \land q) \to r$. We know q is true and r is false. For this implication to hold, $(p \land q)$ must be false (otherwise true $\to$ false would be false). Since q is true, we need p to be false to make $(p \land q)$ false.

Step 5: From (5), $p \lor \neg r$. With p false and r false, we have false $\lor$ true = true. This is satisfied.

Verification: With $p = F, q = T, r = F, s = F, t = F$:

- (1): $(F \land T) \to F = F \to F = T$
- (2): $F \to (F \lor F) = F \to F = T$
- (3): $\neg F \land \neg F = T \land T = T$

- (4): T
- (5): $F \vee \neg F = F \vee T = T$

Evaluation of options:

A) **Incorrect.** KB is satisfiable as shown above.

B) **Incorrect.** KB forces p to be false, not true.

C) **Correct.** KB $\models \neg p$ because p must be false in all models.

D) **Incorrect.** KB forces both r and s to be false, so $(r \wedge s)$ is false.

E) **Incorrect.** If $p = T$ and $r = T$, then from (2) we need $(s \vee t)$ to be true. But (3) forces both s and t to be false. Contradiction.

Key Concept: This problem requires backward reasoning from constraints. Statement (3) forces s and t false, which through (2) forces r false, which through (1) and (4) forces p false. The chain isn't immediately obvious, making this challenging.

Question I.3: Constraint Satisfaction Problems — Arc Consistency

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 2

Consider a CSP with variables A, B, C.
Initial Domains: $D(A) = \{1, 2\}$, $D(B) = \{1\}$, $D(C) = \{1, 2\}$
Constraints:

- $A \neq B$
- $B < C$
- $C \leq 2$ (already reflected in $D(C)$)

Run AC-3 (Arc Consistency) checks and remove impossible values until nothing changes.
Which are true?

A) After AC, $D(C)$ becomes $\{2\}$.

B) After AC, $D(A)$ becomes $\{2\}$.

C) After AC, all domains become singletons and the CSP is solved without search.

D) If we replace "$B < C$" by "$B \leq C$", AC would still make all domains singletons.

E) If we keep "$B < C$" but widen $D(B)$ to $\{1, 2\}$, AC must still prune C to $\{2\}$.

Answer: A, B, C, E

Explanation: Arc consistency propagates constraints to prune domain values that have no support. In this case, AC-3 reduces all domains to singletons, solving the CSP.

Details:

A) **Correct.** Since $D(B) = \{1\}$ and the constraint $B < C$ holds, we need $1 < C$. From $D(C) = \{1, 2\}$, only $C = 2$ satisfies this. Therefore, AC prunes $D(C)$ to $\{2\}$.

B) **Correct.** The constraint $A \neq B$ with $D(B) = \{1\}$ means $A \neq 1$. Since $D(A) = \{1, 2\}$, AC prunes $A = 1$, leaving $D(A) = \{2\}$.

C) **Correct.** After applying AC:

- $D(A) = \{2\}$
- $D(B) = \{1\}$

- $D(C) = \{2\}$

All domains are singletons, and the assignment $A = 2, B = 1, C = 2$ satisfies all constraints. The CSP is solved without backtracking search.

D) **Incorrect.** If we replace "$B < C$" with "$B \leq C$", then with $B = 1$, both $C = 1$ and $C = 2$ satisfy the constraint. AC would not prune $D(C)$ to a singleton; it would remain $\{1, 2\}$. Similarly, this doesn't help prune other domains further.

E) **Correct.** If $D(B) = \{1, 2\}$, the AC-3 algorithm proceeds as follows:

- **Step 1:** Check arc (B, C) for constraint $B < C$:
  - For $B = 1$: C must be > 1, so $C = 2$ is supported.
  - For $B = 2$: C must be > 2, but $D(C) = \{1, 2\}$ has no such value. **Prune** $B = 2$.

  After pruning: $D(B) = \{1\}$
- **Step 2:** Since $D(B)$ changed, re-check arc (C, B) for constraint $B < C$:
  - For $C = 1$: need $B < 1$, but $D(B) = \{1\}$ has no such value. **Prune** $C = 1$.
  - For $C = 2$: need $B < 2$, and $B = 1$ satisfies this. $C = 2$ is supported.

  After pruning: $D(C) = \{2\}$

Final result: AC-3 does indeed prune C to $\{2\}$, making this statement correct. The algorithm may prune B first as an intermediate step, but the final outcome is that $D(C) = \{2\}$, which is exactly what the statement claims.

Key Concept: Arc consistency enforces local consistency by ensuring every value in a variable's domain has a supporting value in the domains of related variables under the constraints.

Question I.4: Logical (Knowledge-Based) Agents — Architecture vs. Environment

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 1, CLO 3

A company deploys **autonomous delivery drones** operating in a complex city environment that is **partially observable** (limited sensors and occasional signal loss), **stochastic** (uncertain wind and traffic conditions), **sequential** (each action affects future deliveries), and **dynamic** (the world keeps changing while the agent is acting).

Evaluate the following statements about possible agent architectures and identify which are correct.

A) A simple reflex agent with a fixed condition–action table can act rationally in such an environment if its rule base is sufficiently large to cover all possible situations the drone may face during operation.

B) A model-based reflex agent can outperform a simple reflex agent in this scenario by maintaining an internal belief about unobserved factors such as wind direction, obstacle positions, or battery status, allowing it to make more informed decisions.

C) A goal-based agent can only behave rationally if the effects of all actions are deterministic and known in advance, since uncertainty in outcomes would prevent it from reliably selecting actions that achieve its goals.

D) A utility-based agent can prefer actions that reduce uncertainty — such as temporarily rising to a higher altitude to gain better GPS visibility — even when these actions do not immediately improve goal satisfaction.

E) In such environments, table-driven agents are always optimal if the table covers every possible percept sequence, regardless of changes or randomness in the environment.

Answer: B, D

Explanation: Model-based agents handle partial observability through internal state, and utility-based agents can rationally value information-gathering actions.

Details:

A) **Incorrect.** A simple reflex agent responds only to the current percept without memory or state. In a partially observable environment, the current percept alone is insufficient to determine the optimal action. In stochastic and dynamic environments, a fixed table cannot feasibly cover all contingencies. Even if "sufficiently large" theoretically means covering all cases, this sidesteps practical infeasibility and doesn't guarantee rationality when the environment changes unpredictably.

B) **Correct.** A model-based reflex agent maintains an **internal state** that represents beliefs about unobserved aspects of the environment. This allows it to make better decisions than a simple reflex agent, which has no memory and cannot reason about hidden state. In partially observable environments, maintaining beliefs is crucial for rational behavior.

C) **Incorrect.** Goal-based agents can be rational in stochastic environments. They can use probabilistic planning, decision theory, and expected utility to handle uncertainty. Stochasticity doesn't preclude rationality—it requires more sophisticated reasoning about probabilities and outcomes.

D) **Correct.** Utility-based agents maximize expected utility, which can include the value of information. In uncertain environments, gathering information (exploration) can have high utility even if it doesn't immediately achieve a goal, because it enables better future decisions. This is a key advantage of utility-based agents over purely goal-based agents.

E) **Incorrect.** Table-driven agents that store responses for all possible percept sequences face several problems:

- In stochastic environments, the same percept sequence may require different actions depending on hidden state or random outcomes.
- In dynamic environments, the optimal policy may change over time.
- The space of all possible percept sequences is typically infinite or exponentially large, making exhaustive tables infeasible.
- Covering all sequences doesn't guarantee optimality if the environment's dynamics or utility function changes.

Key Concept: Agent architecture must match environment properties. Partial observability requires state tracking, stochasticity requires probabilistic reasoning, and dynamic environments require adaptability beyond fixed tables.

Part II. Logical Reasoning in Education Context

Total Score: 10 points **Total Time:** 40 minutes

Given Predicates:

Predicate	Meaning
Student(x)	x is a student
Teacher(x)	x is a teacher
Course(x)	x is a course
Takes(x, y)	Student x takes course y
Teaches(x, y)	Teacher x teaches course y
Understands(x, y)	Student x understands topic y
Topic(y)	y is a topic
Covers(c, y)	Course c covers topic y
GradedBy(x, t)	Student x is graded by Teacher t
Passes(x, y)	Student x passes course y

Translate each of the following statements into first-order logic (FOL) using the provided predicates, then perform the required logical transformation.

Question II.1: Translation to FOL

Time: 10 minutes **Score:** 3 points **Mapped CLOs:** CLO 1

Translate the following statement into first-order logic (FOL):
"Every student who takes a course taught by a teacher understands at least one topic that the course covers."

Answer:
$\forall s, t, c\,(\text{Student}(s) \wedge \text{Teacher}(t) \wedge \text{Course}(c) \wedge \text{Takes}(s, c) \wedge \text{Teaches}(t, c)) \rightarrow$
$(\exists u\,(\text{Topic}(u) \wedge \text{Covers}(c, u) \wedge \text{Understands}(s, u)))$
Explanation:

- The universal quantifiers $\forall s, t, c$ range over all possible students, teachers, and courses.
- The antecedent ensures we're considering a student who takes a course that is taught by a teacher.
- The consequent states there exists at least one topic u that is both covered by the course and understood by the student.

Question II.2: Translation with Negation

Time: 10 minutes **Score:** 3 points **Mapped CLOs:** CLO 1

Translate the following statement into first-order logic (FOL):
"No teacher grades a student who is not enrolled in one of their courses."

Answer:
$\forall t, s\,((\text{Teacher}(t) \wedge \text{Student}(s) \wedge \text{GradedBy}(s, t)) \rightarrow$
$(\exists c\,(\text{Course}(c) \wedge \text{Teaches}(t, c) \wedge \text{Takes}(s, c))))$
Explanation:

- The universal quantifiers $\forall t, s$ range over all teachers and students.
- The antecedent identifies cases where a teacher grades a student.

- The consequent requires that there exists at least one course that the teacher teaches and the student takes.
- This formulation captures the constraint that grading implies enrollment.

Answer (any one of the following is correct):

(A) Implication form (enrollment required to be graded):

$$\forall t\,\forall s\,\big((\text{Teacher}(t) \land \text{Student}(s) \land \text{GradedBy}(s,t)) \to \exists c\,(\text{Course}(c) \land \text{Teaches}(t,c) \land \text{Takes}(s,c))\big).$$

(B) Contrapositive style (closest to "No teacher grades ... not enrolled"):

$$\forall t\,\forall s\,\big(\text{Teacher}(t) \land \text{Student}(s) \land \neg\exists c\,(\text{Course}(c) \land \text{Teaches}(t,c) \land \text{Takes}(s,c)) \to \neg\text{GradedBy}(s,t)\big).$$

(C) Negated-existence of a violation:

$$\neg\exists t\,\exists s\,\big(\text{Teacher}(t) \land \text{Student}(s) \land \text{GradedBy}(s,t) \land \forall c\,((\text{Course}(c) \land \text{Teaches}(t,c)) \to \neg\text{Takes}(s,c))\big).$$

(D) Using a shorthand predicate (if allowed):

$$\text{EnrolledWith}(s,t) \;\equiv\; \exists c\,(\text{Course}(c) \land \text{Teaches}(t,c) \land \text{Takes}(s,c)),$$

$$\forall t\,\forall s\,\big((\text{Teacher}(t) \land \text{Student}(s) \land \text{GradedBy}(s,t)) \to \text{EnrolledWith}(s,t)\big).$$

Explanation (brief):

- (A) expresses that grading implies enrollment in a course taught by that teacher.
- (B) says if the student is not enrolled with that teacher, they cannot be graded by them.
- (C) forbids the existence of any grading event violating the rule.
- (D) uses a shorthand predicate for clarity and equivalence.

Question II.3: Negation and Conjunctive Normal Form (CNF)

Time: 20 minutes **Score:** 4 points **Mapped CLOs:** CLO 1

Negate the following statement and convert it step-by-step into Conjunctive Normal Form (CNF). Show all transformations clearly and justify each step using the appropriate logical rule (e.g., $\neg\forall$, implication removal, De Morgan's law, or Skolemization).

$$\forall x\,(\text{Student}(x) \to \text{Hardworking}(x))$$

Solution Steps:

Step 1: Negate the statement

$$\neg\forall x\,(\text{Student}(x) \to \text{Hardworking}(x))$$

Step 2: Move negation inward

$$\exists x\,\neg(\text{Student}(x) \to \text{Hardworking}(x))$$

Step 3: Replace implication ($A \rightarrow B \equiv \neg A \vee B$)

$$\exists x \, (\text{Student}(x) \wedge \neg \text{Hardworking}(x))$$

Step 4: Skolemization (replace $\exists x$ with constant a)

$$\text{Student}(a) \wedge \neg \text{Hardworking}(a)$$

Final CNF form (conjunction of literals):

$$\text{Student}(a) \wedge \neg \text{Hardworking}(a)$$

Interpretation: There exists at least one student a who is not hardworking.

Part III. Constraint Satisfaction Problem (CSP)

Total Score: 6 points **Total Time:** 30 minutes

Part III: Constraint Satisfaction Problems (CSP)

Problem Specification (Student-Friendly Version)

Context: You are designing the schedule for **AI Lab presentation sessions** on Mondays. There are three labs and three teaching assistants (TAs) available. Each TA can supervise at most one lab at a time, and some labs overlap in time. Your goal is to assign each lab to a TA in a way that satisfies all constraints.

Labs and Meeting Times

Lab	Topic	Time
Lab 1	Search Algorithms	8:00–9:00 am
Lab 2	Machine Learning	8:30–9:30 am
Lab 3	Knowledge Representation	9:00–10:00 am

TA Availability

TA	Available for Labs
TA A	Labs 1 and 2
TA B	Labs 2 and 3
TA C	Labs 1, 2, and 3

Assignment Rules (Constraints)

1. Each lab must be supervised by exactly one TA.
2. A TA cannot supervise overlapping labs (time conflicts).
3. A TA can only be assigned to labs where they are available.

Question III.1: Formulate as a CSP

Time: 10 minutes **Score:** 2 points **Mapped CLOs:** CLO 2

You are asked to formulate the problem of the scheduling scenario as a **Constraint Satisfaction Problem (CSP)**.

Clearly specify:

- the **variables**,
- the **domains** of each variable, and
- the **binary constraints** that must hold between the variables.

You may describe the constraints in English words.

Answer:

Variables (one per lab):

- L_1, L_2, L_3

Domains (based on TA availability):

- $D(L_1) = \{A, C\}$
- $D(L_2) = \{A, B, C\}$

- $D(L_3) = \{B, C\}$

Constraints (overlapping sessions cannot share the same TA):

- $L_1 \neq L_2$ (Labs 1 and 2 overlap during 8:30–9:00am)
- $L_2 \neq L_3$ (Labs 2 and 3 overlap during 9:00–9:30am)

Note: Lab 1 and Lab 3 do not overlap (Lab 1 ends at 9:00, Lab 3 starts at 9:00), so no constraint is needed between them.

Grading Rubric (2 points):

- **0.5 pt:** Correct variables definition
- **1.0 pt:** Correct domains based on TA availability
- **0.5 pt:** Correct constraints identifying overlapping sessions

Question III.2: Arc Consistency (AC-3)

Time: 15 minutes **Score:** 3 points **Mapped CLOs:** CLO 2

Using the CSP formulation from Question III.1, apply the **Arc Consistency (AC-3)** algorithm.

- Check each relevant arc between lab variables based on the binary constraints.
- For each value in a variable's domain, determine whether it has at least one supporting value in the connected variable's domain.
- Indicate which (if any) domain values are removed and explain why.
- Continue until a **fixed point** is reached (no further pruning possible).
- State the **final domains** for all variables after enforcing arc consistency.

Hint: For constraints of the form $L_i \neq L_j$, a value x in $D(L_i)$ is supported if there exists at least one y in $D(L_j)$ such that $x \neq y$.

Answer:

Initial domains:

- $D(L_1) = \{A, C\}$
- $D(L_2) = \{A, B, C\}$
- $D(L_3) = \{B, C\}$

Arc Consistency Process:

Step 1: Check arc (L_2, L_1) for constraint $L_1 \neq L_2$

- For each value in $D(L_2)$, check if there exists a supporting value in $D(L_1)$:
  - $L_2 = A$: L_1 can be C (since $A \neq C$). **Supported.**
  - $L_2 = B$: L_1 can be A or C (since $B \neq A$ and $B \neq C$). **Supported.**
  - $L_2 = C$: L_1 can be A (since $C \neq A$). **Supported.**
- **No pruning** from $D(L_2)$.

Step 2: Check arc (L_1, L_2) for constraint $L_1 \neq L_2$

- For each value in $D(L_1)$, check if there exists a supporting value in $D(L_2)$:

- $L_1 = A$: L_2 can be B or C (since $A \neq B$ and $A \neq C$). **Supported.**
- $L_1 = C$: L_2 can be A or B (since $C \neq A$ and $C \neq B$). **Supported.**

- **No pruning** from $D(L_1)$.

Step 3: Check arc (L_2, L_3) for constraint $L_2 \neq L_3$

- For each value in $D(L_2) = \{A, B, C\}$, check support in $D(L_3) = \{B, C\}$:
  - $L_2 = A$: L_3 can be B or C (since $A \neq B$ and $A \neq C$). **Supported.**
  - $L_2 = B$: L_3 can be C (since $B \neq C$). **Supported.**
  - $L_2 = C$: L_3 can be B (since $C \neq B$). **Supported.**
- **No pruning** from $D(L_2)$.

Step 4: Check arc (L_3, L_2) for constraint $L_2 \neq L_3$

- For each value in $D(L_3) = \{B, C\}$, check support in $D(L_2) = \{A, B, C\}$:
  - $L_3 = B$: L_2 can be A or C (since $A \neq B$ and $C \neq B$). **Supported.**
  - $L_3 = C$: L_2 can be A or B (since $A \neq C$ and $B \neq C$). **Supported.**
- **No pruning** from $D(L_3)$.

Final Arc-Consistent Domains:

- $D(L_1) = \{A, C\}$ *(unchanged)*
- $D(L_2) = \{A, B, C\}$ *(unchanged)*
- $D(L_3) = \{B, C\}$ *(unchanged)*

Conclusion: Arc consistency does not prune any values in this CSP. All initial domain values have support. The CSP is arc-consistent but not solved—backtracking search is still required.

Grading Rubric (3 points):

- **1.5 pts:** Correct identification that no pruning occurs
- **1.0 pt:** Showing work (checking at least 2 arcs systematically)
- **0.5 pt:** Correct final domains stated clearly

Partial Credit:

- If student incorrectly prunes values but shows logical reasoning: 0.5–1.5 pts
- If student only states final domains without work: max 0.5 pts

Question III.3: Find a Consistent Solution

Time: 5 minutes **Score:** 1 point **Mapped CLOs:** CLO 2

Give **one valid assignment** of TAs to labs that satisfies all constraints.

Answer:
Multiple valid solutions exist. Any of the following is acceptable:
Solution 1:

- $L_1 = A$

- $L_2 = B$
- $L_3 = C$

Solution 2:

- $L_1 = A$
- $L_2 = C$
- $L_3 = B$

Solution 3:

- $L_1 = C$
- $L_2 = A$
- $L_3 = B$

Solution 4:

- $L_1 = C$
- $L_2 = B$
- $L_3 = C$

Verification (for Solution 1):

- Domain check: $A \in D(L_1)$, $B \in D(L_2)$, $C \in D(L_3)$
- Constraint $L_1 \neq L_2$: $A \neq B$
- Constraint $L_2 \neq L_3$: $B \neq C$

All constraints satisfied.

Grading Rubric (1 point):

- **1.0 pt:** Any valid assignment that satisfies all constraints
- **0.0 pt:** Invalid assignment (violates constraints or domains)

Note: Students do not need to verify the solution explicitly—a correct assignment is sufficient.

Answer Key Summary:

- Question III.1: Variables: L_1, L_2, L_3; Domains: $\{A, C\}, \{A, B, C\}, \{B, C\}$; Constraints: $L_1 \neq L_2$, $L_2 \neq L_3$
- Question III.2: All domains remain unchanged (no pruning)
- Question III.3: Any valid assignment (e.g., $L_1 = A, L_2 = B, L_3 = C$)